

Pharmaceutical Reform and Pharmacy Entry

County-level Results, v1: Original Policy-Year Timing

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Empirical Setting

- ▶ Unit of analysis: county-by-year panel.
- ▶ Controlled sample: 2012–2019, with population, nighttime lights, and road density.
- ▶ v1 timing uses the original policy year:

$$Reform_{ct} = \mathbf{1}\{t \geq PolicyYear_c\}.$$

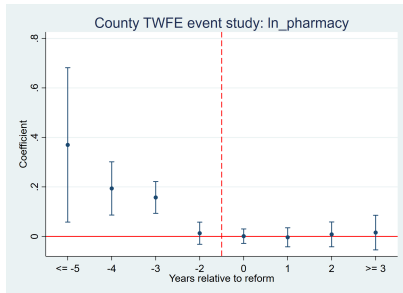
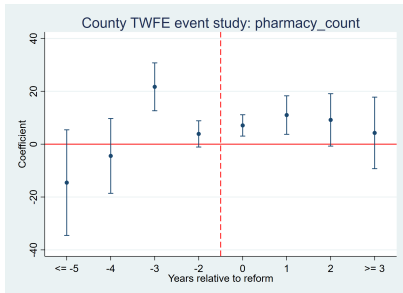
- ▶ Standard errors are clustered at the county level.
- ▶ TWFE and CSDID event-study figures use the same outcome definitions and event-time window as the five-estimator robustness figures.

TWFE Baseline Results: Count and ln(count)

	Count	ln(count)
Post policy	5.449*** (1.454)	0.020 (0.014)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	16,961	15,469
R ²	0.845	0.882

Controls include log population, log nighttime lights, and log road density.

TWFE Event-study Results



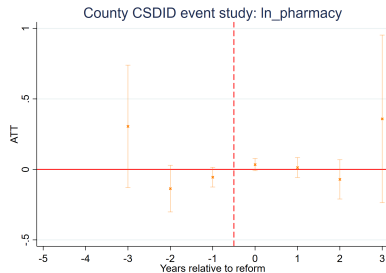
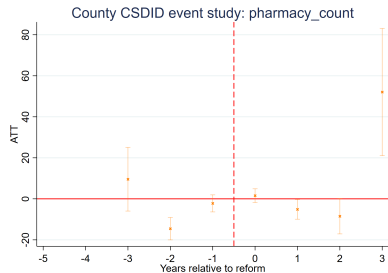
Relative year -1 is omitted. County and year fixed effects are absorbed.

CSDID Results

	Count	ln(count)
ATT	-2.178 (2.077)	0.010 (0.032)
Estimator	CSDID	CSDID
Controls	Yes	Yes
Control group	Not-yet-treated	Not-yet-treated

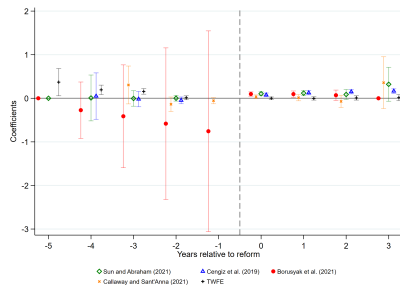
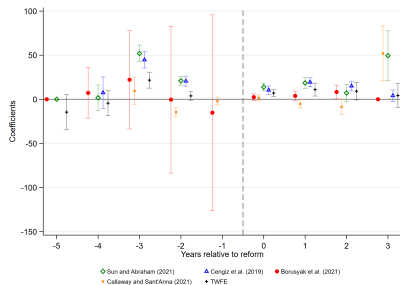
CSDID uses Callaway–Sant’Anna group-time ATT with doubly robust inverse probability weighting and county-clustered standard errors.

CSDID Event-study Results



Same event window as the TWFE and five-estimator event-study figures; relative year -1 is omitted.

Five-estimator Event-study Robustness



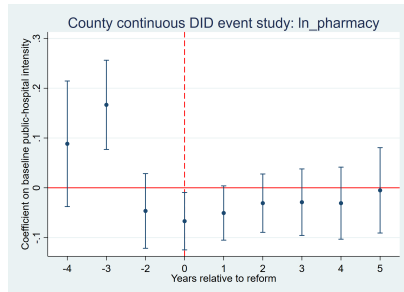
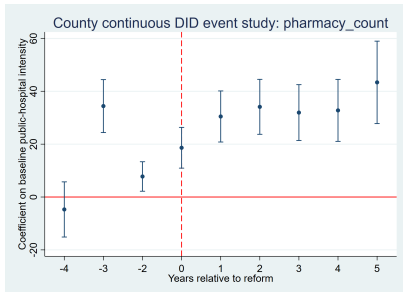
Estimators: TWFE, Callaway–Sant’Anna CSDID, Sun–Abraham interaction-weighted, Borusyak–Jaravel–Spiess imputation, and stacked event study.

Continuous DID: OLS and IV Baseline Results

	OLS		IV: 1959 stock	
	Count	ln(count)	Count	ln(count)
Post $\times$ ln(public hospitals)	20.672*** (3.776)	-0.068** (0.024)	99.574*** (15.529)	0.162 (0.097)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	16,961	15,469	13,788	12,463
R ² / K-P F	0.847	0.882	81.389	135.751

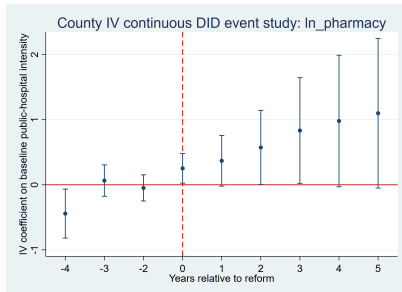
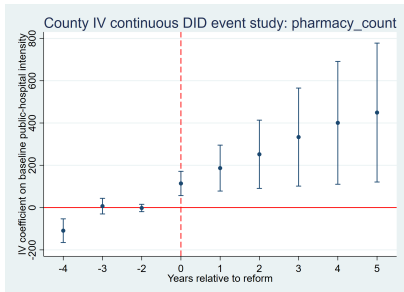
IV instruments interact post policy with the 1959 epidemiology-station stock.

Continuous DID Event-study: OLS



Event-time indicators are interacted with baseline public-hospital intensity.

Continuous DID Event-study: IV



Dynamic IV first stages are weak, so these figures are diagnostic rather than the main IV evidence.

Takeaways

- ▶ TWFE suggests a positive count effect under original policy timing, while the log-count effect is small.
- ▶ CSDID static ATT is not positive for count or log count, so staggered-timing robustness weakens the TWFE interpretation.
- ▶ Five-estimator event-study figures place TWFE and CSDID in the same event-time comparison set.
- ▶ Continuous DID OLS and static IV both imply larger count effects in counties with more baseline public hospitals.
- ▶ IV event-study plots are retained as diagnostics; the main IV evidence is the static continuous-DID specification.